Md. Rafiul Islam

Computer Science and Engineering Student
Dhaka, Bangladesh | 01845904415 | rafit991@gmail.com
linkedin.com/in/rafitboo | github.com/rafitboo

PROFILE STATEMENT

Computer Science and Engineering student with hands-on experience across web development, machine learning, and computer graphics. Track record of building full-stack applications with Flask and Django, implementing 3D rendering systems with PyOpenGL, and developing federated learning architectures for real-world classification tasks. Comfortable owning a project end to end, from system design through deployment and presentation.

SKILL SETS

- **Languages & Web:** Python, C/C++, JavaScript, HTML/CSS, Flask, Django
- **Databases & Tools:** MySQL, SQLAlchemy, Pytest, Git, PyOpenGL, GLUT
- **Machine Learning:** Federated Learning, Variational Autoencoders (VAEs), Dirichlet Partitioning, Hierarchical Models
- **Networking & OS:** Cisco Routing, VLSM, Fedora Linux, Windows

PROJECTS

Federated Learning with Hierarchical Buffer Clustering (Thesis) **July '25 – June '26**
Python, PyTorch, Federated Learning

- Developed a federated learning framework across six heterogeneous clients sharing a common classifier head via FedAvg with FedProx regularization.
- Designed a hierarchical clustering-based buffer (HAC on ConvNeXt features) that improved non-buffer class accuracy by 12.57 percentage points, reaching 63.76% overall test accuracy.
- Validated privacy properties of the framework, measuring 0% property-inference attack accuracy.
- Co-authoring an IEEE conference paper and an Elsevier journal article based on this work.

Hybrid Music Clustering via VAEs | Code
Python, PyTorch, Machine Learning, Unsupervised Learning

- Engineered a deep generative pipeline for a CSE425 Neural Networks project utilizing Basic, Convolutional Hybrid, and Conditional Variational Autoencoders (CVAEs) to cluster music tracks.
- Fused multimodal data streams, successfully transitioning from uni-modal audio feature models (MFCCs) to complex multi-modal representations combining audio and lyric data (TF-IDF).
- Evaluated latent space structures across K-Means, Agglomerative Clustering, and DBSCAN algorithms, utilizing genre-labeled CVAEs to achieve optimal structural separation and semantic accuracy (Purity, NMI).

Medicare: Online Pharmacy Platform | Code
Python, Flask, MySQL, SQLAlchemy, JavaScript

- Architected a multi-role online pharmacy platform (admin, staff, customer) with Python, Flask, and MySQL, following MVC design principles.
- Built secure authentication, role-based dashboards, and a full e-commerce flow with real-time cart management and bKash payment integration.
- Added real-time customer support chat and a Pytest-based backend test suite to ensure system reliability.

ZK Network: Secure Messaging Platform | Code
Python, Django, Cryptography, SQLite

- Developed an end-to-end encrypted messaging platform for a CSE447 Cryptography course, implementing hybrid asymmetric cryptography (RSA, ECC) and server-trusted key escrow from scratch.
- Secured data at rest with zero-trust SQLite storage, enforcing strict 2FA (Email OTP), stateless JWT sessions, and HMAC-SHA256 for message integrity against database tampering.
- Engineered a custom Django management command for a secure master key rotation protocol, actively debugging and resolving associated errors to ensure seamless, atomic data re-encryption.

TRAIM: 3D Aim Trainer & Graphics Algorithms | Code
Python, PyOpenGL, GLUT, 3D Rendering

- Built a 3D aim trainer with multiple game modes (reaction, timed, practice), configurable settings, and detailed performance tracking using Python and PyOpenGL.
- Implemented first-person 3D mechanics including weapon-specific logic (sniper scopes, shotgun spreads), bullet tracers, and hit-impact effects.
- Programmed core computer graphics components, including a 2D physics collision system and line rendering via the midpoint line algorithm.

EXPERIENCE

Student Tutor (ST) — Undergraduate Teaching Assistant **Spring 2026 – Present**
BRAC University *Dhaka, Bangladesh*

- Supported instruction for CSE250: Circuits and Electronics, assisting students with core concepts, problem sets, and lab work.
- Held office hours and review sessions to reinforce course material and improve student outcomes.

ACHIEVEMENTS

- Awarded a merit-based scholarship at BRAC University for academic excellence.
- Solved over **200+ problems** on Codeforces and LeetCode.

EDUCATION

BRAC University **Sep '26 (Expected)**
Bachelor of Science in Computer Science and Engineering, CGPA: 3.93 out of 4.00 *Dhaka, Bangladesh*

TEAMWORK & COLLABORATION

- Collaborated within a cross-functional academic project team, coordinating shared deliverables and rebalancing workload across members to meet deadlines.
- Prepared network configuration overviews and routing methodology presentations for project vivas and academic showcases.